Practical No 02

a) Develop an Android application to create and integrate a Fragment and demonstrate Fragment lifecycle methods.

Project Name: FragmentLifecycleDemo

Create a new fragment class:

java->new-> fragment->fragment(blank)

Fragment Name: DemoFragment

Fragment Layout name: fragment_demo

Program Code:

I. Filename: DemoFragment.java

```
package com.example.fragmentlifecycledemo;

import android.content.Context;

import android.os.Bundle;

import android.util.Log;

import android.view.LayoutInflater; import android.view.View;

import android.view.ViewGroup;

import androidx.annotation.NonNull; import androidx.annotation.Nullable;

import androidx.fragment.app.Fragment;

public class DemoFragment extends Fragment {

   String TAG = "FragmentLifecycle";


   @Override

   public void onAttach(@NonNull Context context) {

   super.onAttach(context);

   Log.d(TAG, "onAttach() executed");}


   @Override

  public void onCreate(Bundle savedInstanceState) {

       super.onCreate(savedInstanceState);

       Log.d(TAG, "onCreate() executed");
```

```
    }


    @Override

    public View onCreateView(LayoutInflater inflater, ViewGroup container,Bundle
savedInstanceState) {

        Log.d(TAG, "onCreateView() executed");

        return inflater.inflate(R.layout.fragment_demo, container, false);

    }


@Override

public void onViewCreated(@NonNull View view, @Nullable Bundle savedInstanceState)
{

        super.onViewCreated(view, savedInstanceState);

         Log.d("on view created invoked", "onViewCreated() executed");

}


    @Override

    public void onStart() {

        super.onStart();

        Log.d("on start invoked", "on started");


    }


    @Override

    public void onResume() {

        super.onResume();

        Log.d("on resume invoked", "on resume");

    }


    @Override

    public void onPause() {

        super.onPause();
```

```
        Log.d("on pause invoked", "on pause");
    }

    @Override
    public void onStop() {
        super.onStop();
        Log.d("on stop invoked", "on stopped");
    }

    @Override
    public void onDestroyView() {
        super.onDestroyView();
        Log.d("on destroyview invoked", "on destroyview() executed");
    }

    @Override
    public void onDestroy() {
        super.onDestroy();
        Log.d("on destroy invoked", "on destroy");
    }

    @Override
    public void onDetach() {
        super.onDetach();
        Log.d("on detach invoked", "on detach");
    }
}
```

II. Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/fragment_container_view_tag"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World!"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

</androidx.constraintlayout.widget.ConstraintLayout>
```

III. Filename: MainActivity.java

```
package com.example.fragmentlifecycledemo;

import android.os.Bundle;
import androidx.fragment.app.FragmentTransaction;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_main);

                DemoFragment fragment = new DemoFragment();

                FragmentTransaction transaction =

                        getSupportFragmentManager().beginTransaction();

                transaction.replace(R.id.fragment_container_view_tag, fragment);

                transaction.commit();

        }

    }
```

IV. Filename: fragment_demo.xml

```
<?xml version="1.0" encoding="utf-8"?>

<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:tools="http://schemas.android.com/tools"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    tools:context=".DemoFragment">


    <!-- TODO: Update blank fragment layout -->

    <TextView

        android:layout_width="match_parent"

        android:layout_height="match_parent"

        android:text="@string/hello_blank_fragment" />


</FrameLayout>
```

b) Develop an Android application implementing Fragment-based UI navigation between multiple screens.

Activity Life Cycle, Activity methods, Multiple Activities, Life Cycle of fragments and multiple fragments.

Project Name: Fragment Lifecycle

Template: Empty Views Activity

Language: Java

Project Architecture: Total 6 files

Program Code:

I .Filename: MainActivity.java

```
package com.example.fragmentlifecycle;

 import androidx.appcompat.app.AppCompatActivity;
 import android.os.Bundle;
 import android.widget.Button;

 public class MainActivity extends AppCompatActivity {

       private boolean isFragmentOneLoaded = true;
       private final androidx.fragment.app.FragmentManager fragmentManager =
getSupportFragmentManager();

       @Override
       protected void onCreate(Bundle savedInstanceState) {
       super.onCreate(savedInstanceState);
       setContentView(R.layout.activity_main);

         Button btnChange = findViewById(R.id.btn_change);  // Ensure ID matches
activity_main.xml
       loadFragmentOne();

       btnChange.setOnClickListener(v -> {
              if (isFragmentOneLoaded) {
              loadFragmentTwo();
              } else {
              loadFragmentOne();
              }
       });
       }

       private void loadFragmentOne() {
          androidx.fragment.app.FragmentTransaction transaction =
fragmentManager.beginTransaction();
       FragmentOne fragment = new FragmentOne();
       transaction.replace(R.id.fragment_holder, fragment);  // Ensure ID matches
activity_main.xml
       transaction.addToBackStack(null);
       transaction.commit();
       isFragmentOneLoaded = true;
       }

       private void loadFragmentTwo() {
          androidx.fragment.app.FragmentTransaction transaction =
```

```
fragmentManager.beginTransaction();
        FragmentTwo fragment = new FragmentTwo();
        transaction.replace(R.id.fragment_holder, fragment);  // Ensure ID matches
activity_main.xml
        transaction.addToBackStack(null);
        transaction.commit();
        isFragmentOneLoaded = false;
        }
 }
```

II. Filename: FragmentOne.java

```
package com.example.fragmentlifecycle;
import android.content.Context;
import android.os.Bundle;
import android.util.Log;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.fragment.app.Fragment;
public class FragmentOne extends Fragment {
        private static final String TAG = "FragmentOne";
        @Override
        public void onAttach(Context context) {
        super.onAttach(context);
        Log.d(TAG, "onAttach");
        }
        @Override
        public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle savedInstanceState) {
        Log.d(TAG, "onCreateView");
        return inflater.inflate(R.layout.fragment_one, container, false);
        }

}
```

III. Filename: FragmentTwo.java

```
package com.example.fragmentlifecycle;
import android.content.Context;
import android.os.Bundle;
import android.util.Log;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.fragment.app.Fragment;
public class FragmentTwo extends Fragment {
        private static final String TAG = "FragmentTwo";
        @Override
        public void onAttach(Context context) {
        super.onAttach(context);
    Log.d(TAG, "onAttach");
        }\
        @Override
        public View onCreateView(LayoutInflater inflater, ViewGroup container, Bundle savedInstanceState) {
        Log.d(TAG, "onCreateView");
        return inflater.inflate(R.layout.fragment_two, container, false);
        }
}
```

IV. Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
        xmlns:tools="http://schemas.android.com/tools"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:orientation="vertical"
        tools:context=".MainActivity">
        <!-- Button with ID btn_change -->
        <!-- FrameLayout with ID fragment_holder -->
        <FrameLayout
        android:id="@+id/fragment_holder"
        android:layout_width="match_parent"
        android:layout_height="0dp"
        android:layout_weight="2">
        <Button
        android:id="@+id/btn_change"
        android:layout_width="365dp"
        android:layout_height="88dp"
        android:layout_margin="15dp"
        android:layout_marginTop="15dp"
        android:backgroundTint="@android:color/holo_green_dark"
        android:text="Change Fragment"
        android:textAllCaps="false"
        android:textSize="24sp" />
        </FrameLayout>
</LinearLayout>
```

V. Filename: fragment_one.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="match_parent">
        <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerInParent="true"
        android:text="Fragment One"
        android:textSize="24sp" />
</RelativeLayout>
```

VI. Filename: fragment_two.xml

```
<?xml version="1.0" encoding="utf-8"?>

<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
        android:layout_width="match_parent"
        android:layout_height="match_parent">
        <TextView
        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_centerInParent="true"
        android:text="Fragment Two"
        android:textSize="24sp" />
</RelativeLayout>
```

c) Develop an Android application demonstrating communication between Activity and Fragment.

Project Name: ActivityFragmentCommunication

Creating Fragment
java package → New→ Fragment → Fragment (Blank)
Fragment Name: DemoFragment
Layout Name: fragment_demo

Program Code:

Filename: activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<FrameLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:id="@+id/fragment_container"

   android:layout_width="match_parent"

   android:layout_height="match_parent"/>
```

Open From res/layout:
II. Filename: fragment_demo.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout

   xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:gravity="center"

   android:orientation="vertical"

   android:padding="20dp">



   <TextView

       android:id="@+id/txtMessage"

       android:layout_width="wrap_content" android:layout_height="wrap_content"
android:textSize="22sp"
```

```
        android:text="Message from Activity"/>


    <Button
        android:id="@+id/btnSend"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Message to Activity"
        android:layout_marginTop="20dp"/>
</LinearLayout>
```

Modify DemoFragment.java.

III. Filename: DemoFragment.java

```
package com.example.activityfragmentcommunication;


import android.content.Context;
import android.os.Bundle;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup; import android.widget.Button;
import android.widget.TextView;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;
public class DemoFragment extends Fragment
{
    TextView txtMessage;
    Button btnSend;
    String activityMessage;


    FragmentCommunication communication;
public interface FragmentCommunication {
    void onMessageFromFragment(String message);
```

```
}


// Receive Activity object
@Override
public void onAttach(@NonNull Context context) {
    super.onAttach(context);
    communication = (FragmentCommunication) context;
}




// Method to receive data from Activity
public static DemoFragment newInstance(String message) {
    DemoFragment fragment = new DemoFragment();
    Bundle bundle = new Bundle();
    bundle.putString("msg", message);
    fragment.setArguments(bundle);
    return fragment;
}



@Override
public View onCreateView(LayoutInflater inflater,ViewGroup container,Bundle
savedInstanceState) {
    View view = inflater.inflate(R.layout.fragment_demo,container, false);
    txtMessage = view.findViewById(R.id.txtMessage);
    btnSend = view.findViewById(R.id.btnSend);
// Receiving data from Activity
    if(getArguments() != null){
        activityMessage = getArguments().getString("msg");
        txtMessage.setText(activityMessage);
    }
```

```
// Sending data to Activity
   btnSend.setOnClickListener(v -> {
        communication.onMessageFromFragment(
                "Hello Activity! Message from Fragment");
   }
   );

        return view;
}
}
```

IV. Filename: MainActivity.java

```
package com.example.activityfragmentcommunication;


import android.os.Bundle;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity
        implements DemoFragment.FragmentCommunication {

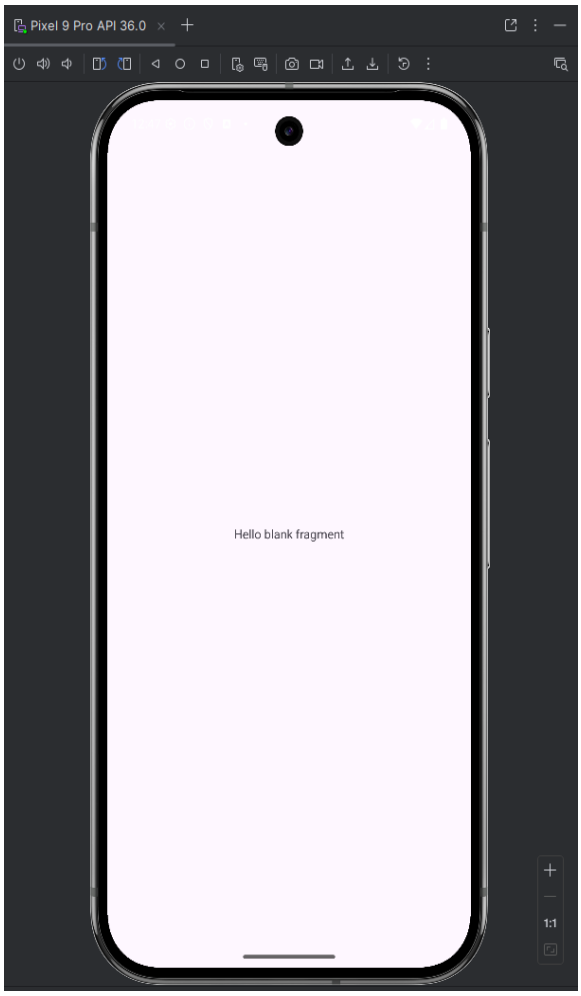

   @Override
   protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);


        // Sending data from Activity to Fragment
        DemoFragment fragment = DemoFragment.newInstance(
                "Hello Fragment! Message from Activity"
        );
```

```
        getSupportFragmentManager()
                .beginTransaction()
                .replace(R.id.fragment_container, fragment)
                .commit();
    }

    // Receiving data from Fragment
    @Override
    public void onMessageFromFragment(String message) {
        Toast.makeText(this, message, Toast.LENGTH_LONG).show();
    }
}
```

a) **Output:**

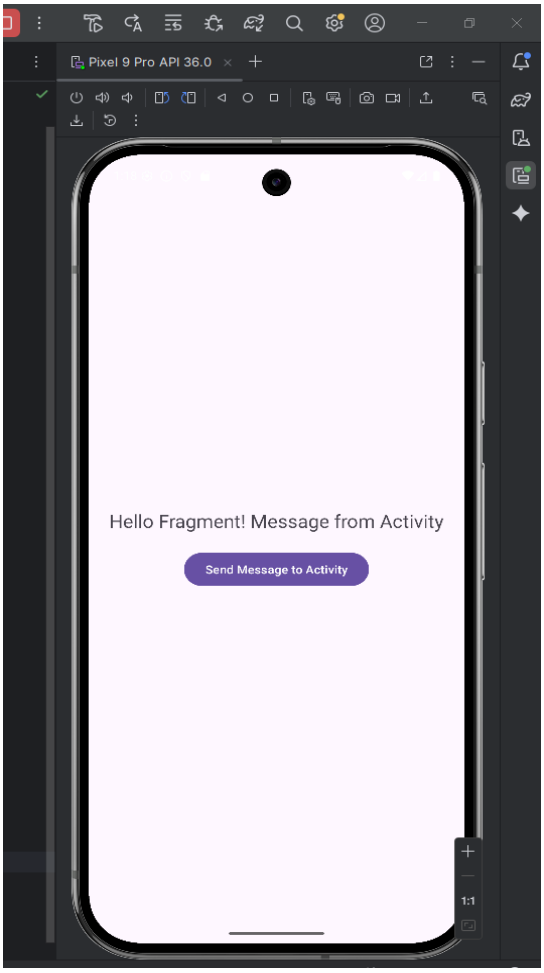


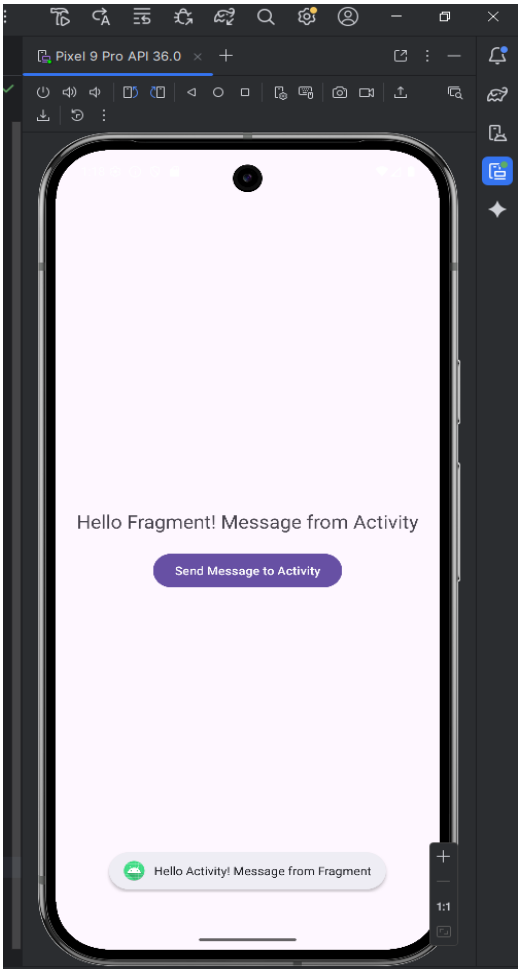
b) Output:


Pixel 9 Pro API 36.0
Change Fragment
Fragment One


Pixel 9 Pro API 36.0
Change Fragment
Fragment Two

c) **Output:**


Pixel 9 Pro API 36.0
Hello Fragment! Message from Activity
Send Message to Activity
1:1


Pixel 9 Pro API 36.0
Hello Fragment! Message from Activity
Send Message to Activity
Hello Activity! Message from Fragment
1:1